



Market Briefs **Pro**

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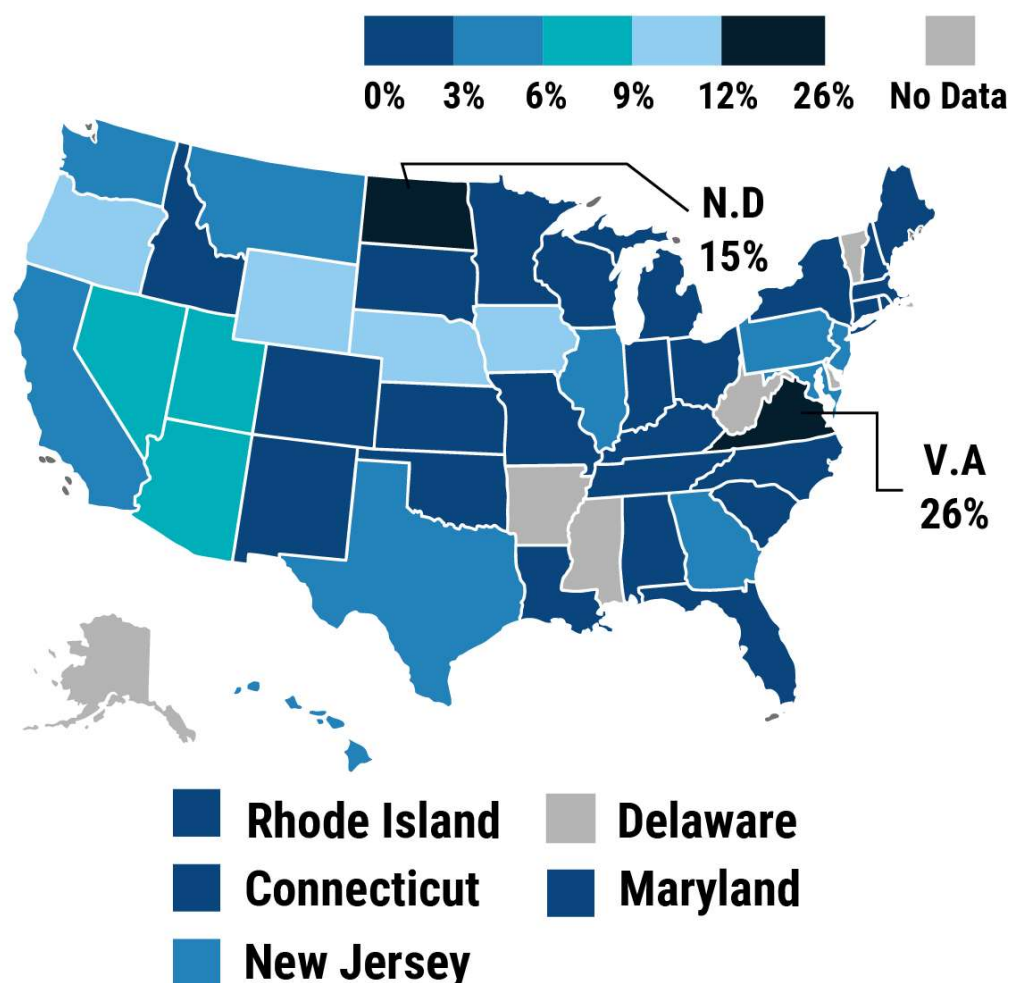
Good morning, Pro Briefers!

This week we've identified an Innovation Shift: AI demand is creating an energy shortage for data centers, causing energy companies to shift their spending into new energy solutions.

Let's break down how investors can take note of this opportunity.

The energy needed to operate data centers is growing - by 2030, Barclays estimates that 10% of the world's energy will be used to power data centers.

Data Centers' Share of Total Power Consumption, 2023



- It takes 2.9 watt-hours of energy to use ChatGPT one time - that's 10X the energy required for a single Google Search.

ChatGPT and other AI tools are powered by data centers.

The problem? Data centers are running low on energy.

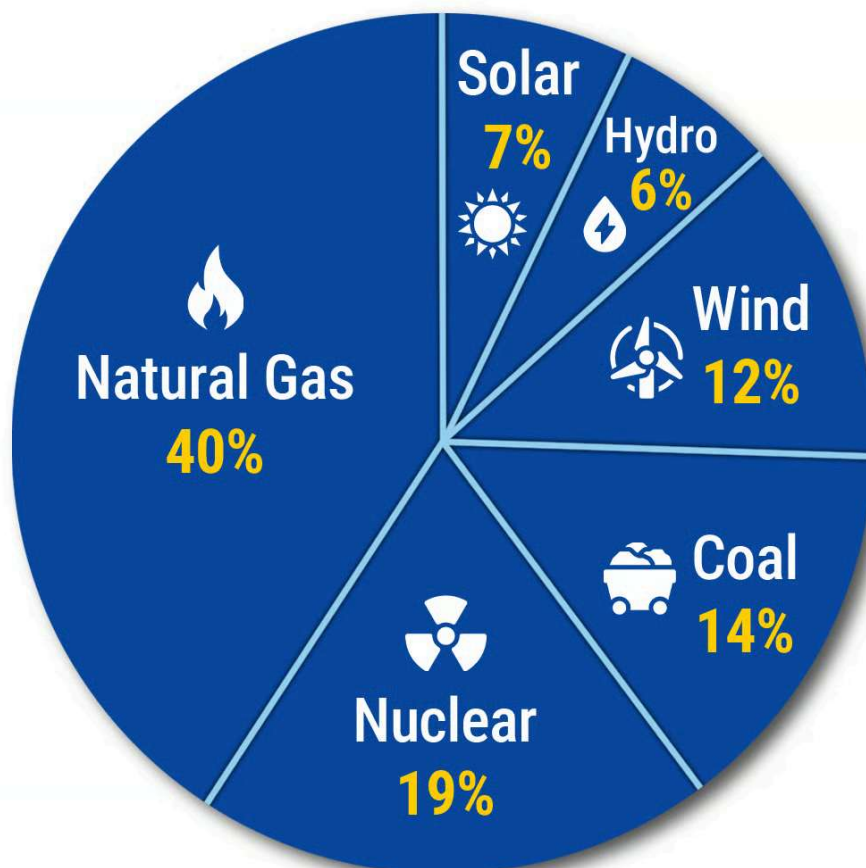
- Data centers globally use more energy than several countries combined - like Italy, Taiwan and Australia.

Researcher Data Center Knowledge anticipates that by 2027 40% of data centers globally will be constrained by power availability.

Why does it matter? President Trump's pick for energy secretary and the Interior Department have both stressed that the U.S. will need more electricity to lead the global AI race.

- President Biden signed an executive order that enables the Department of Energy to help fund data center infrastructure projects.

U.S. Electricity Generation By Source: 2024



This has created an Innovation Shift in spending for energy companies - as power management, nuclear and wind energy businesses are adapting to meet demand.

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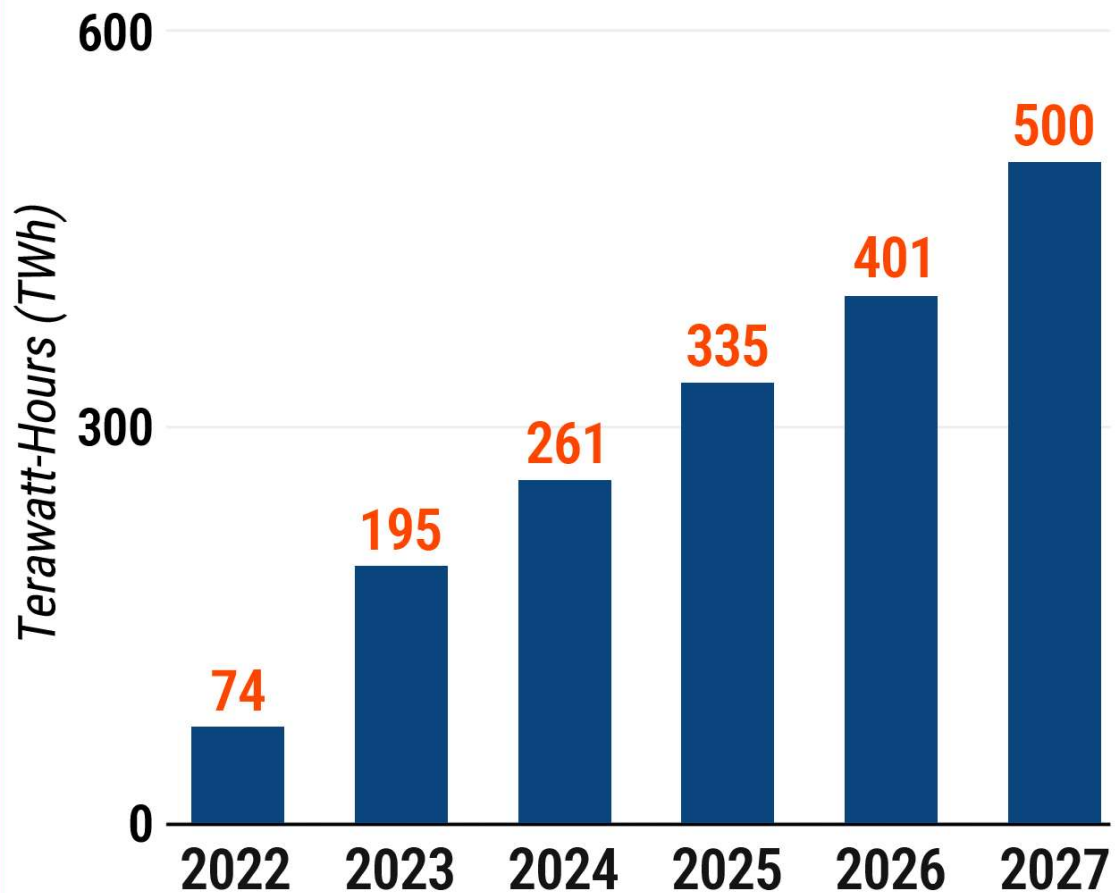
BRIEF BREAKDOWN

Power Management

The average U.S. data center uses more power in a single day than 20,000 U.S. households will use in a year.

This amount of energy puts more stress on our power grid.

Projected Energy Demand For AI Data Centers (2022–2027)



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Data [via](#) Gartner

Without updates to the grid, the International Data Corporation estimates that energy costs could rise 15% a year in the regions where data centers are located.

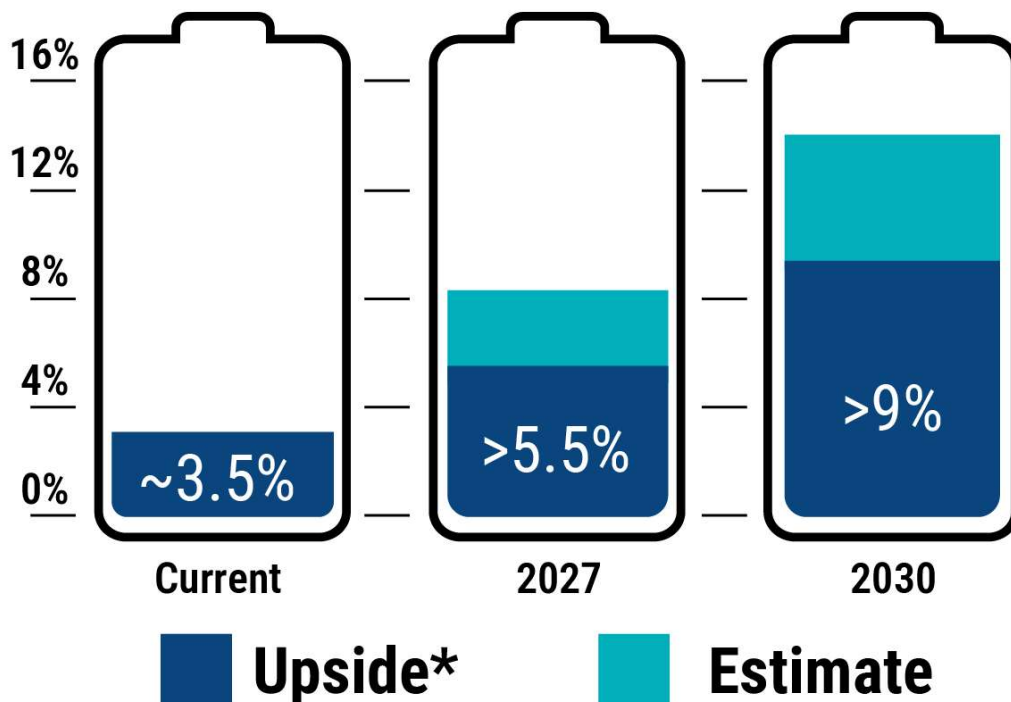
- Power grid failures account for half of all data center downtime, and cost roughly \$100,000 per incident per data center.

Instead of altering our entire power grid with new tech, companies are connecting data centers to the grid with semiconductor wires.

These semiconductor wires give data centers energy at a lower voltage.

In English? Lower voltage cables = higher energy efficiency.

Data Center Demand As % Of Electricity Generation



Based On 2023 Total Net Generation Of 4.178 TWh

**Upside Refers To The Amount Of Energy That Can Be Stored In Non-Peak Hours*

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Data [via](#) the EIA

American Superconductor (AMSC) is one of the leading semiconductor producers in the U.S. and has been designing high-efficiency semiconductor cables for data centers since 2016.

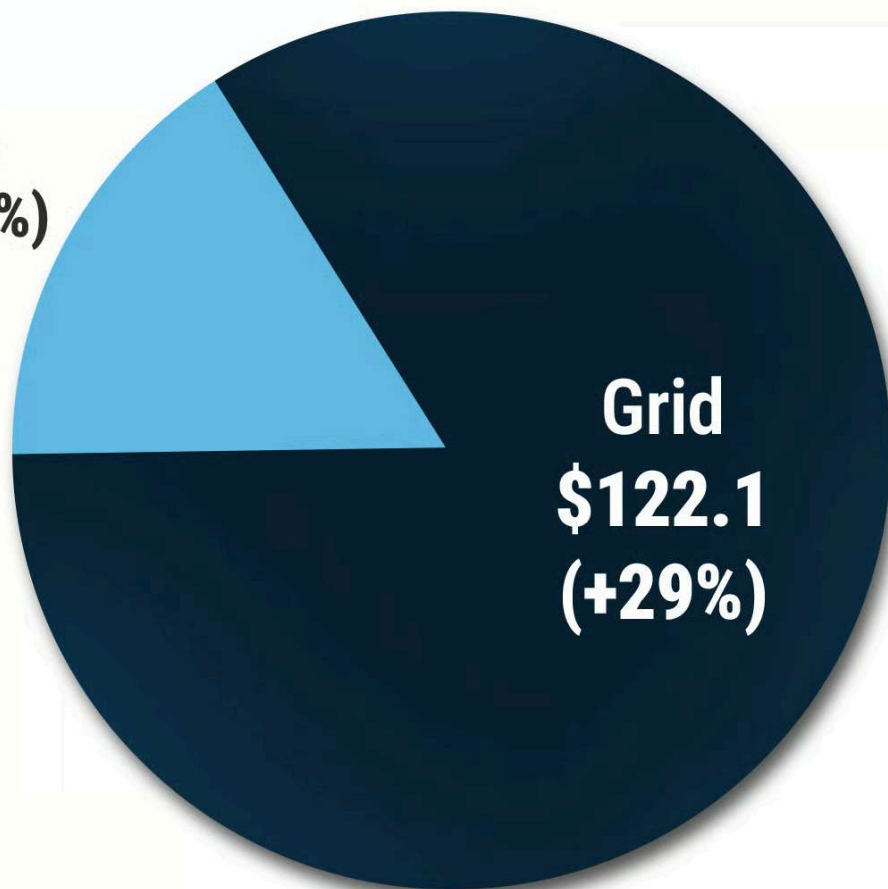
According to the company, its wires can provide up to 10 times more power to data centers - making them the most powerful in the space.

In addition, sales for its semiconductor wires rose 60% year over year in 2024 and 84% of its entire business is in grid solutions.

American Superconductor Revenue Breakdown (2024)

2024 Revenue, In Millions, With Percent Gain From 2023

Wind
\$23.6
(+107.7%)



Grid
\$122.1
(+29%)

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Data [via](#) American Superconductor

- The company also ended last quarter with \$300 million in active contracts.

Those contracts are set to be paid out over the next 5-10 years.

And over the last several years, the company has added around \$40 million in orders for its semiconductor wires.

- In total, the company brought in \$145 million in revenue in 2023, a 37% increase year over year.

In short - American Superconductor holds some of the highest-valued contracts in the data center powering space and its revenue should grow alongside data center power demand.

American Superconductor Corporation (AMSC)

5 Year Price →

32.85
(+364.64%)



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Data [via](#) Yahoo! Finance

Eaton Corporation (ETN) also provides power solutions for data centers - this time in the form of power distribution systems.

These systems act as generators that convert high-voltage power from the grid into low-voltage power a data center can actually use.

Most of Eaton's business is wrapped up in auto manufacturing and traditional grid infrastructure.

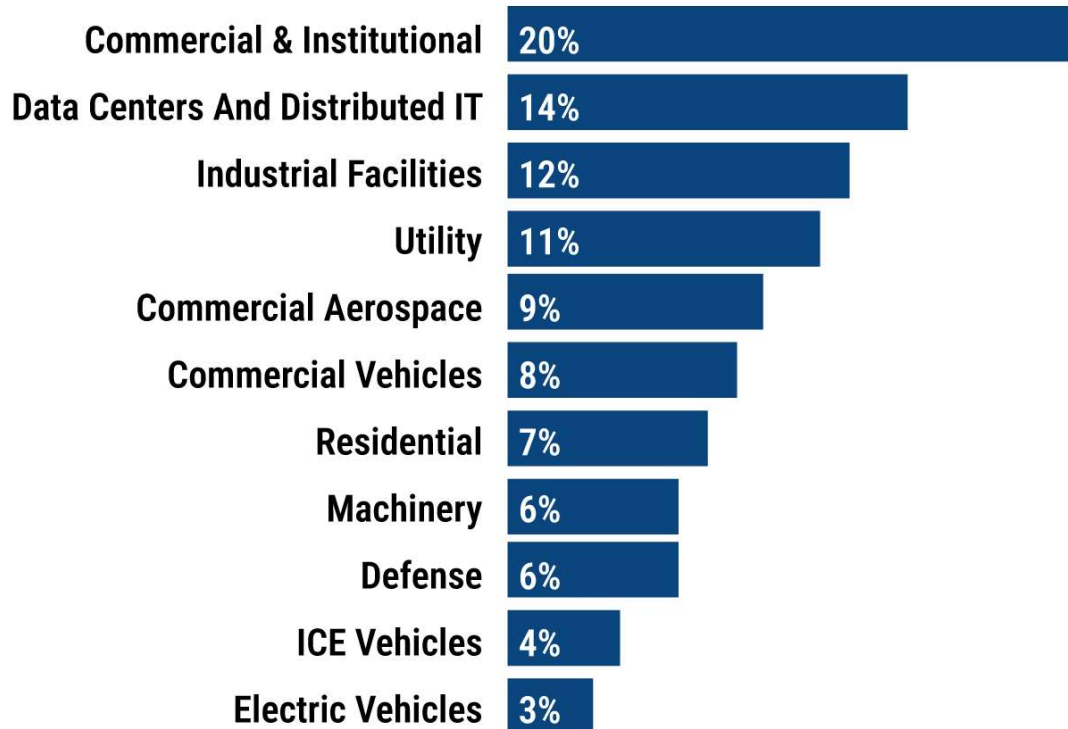
As a result, it anticipated reporting a profit loss for 2024, due to lower demand for auto sales and natural disasters.

However, the company managed to increase profits by 13%.

It's Electric

Data Centers Have Become One Of Eaton's Most Important Markets

■ Market As % Of 2023 Sales



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Data [via](#) Bloomberg

How? Its data center sales rose by 14% year over year - fueled by data center demand rising alongside power needs.

That brought its data center revenue to over \$2 billion for the year - an all-time record.

Eaton's power distribution system innovation can help data centers directly control their own electricity - putting it in high demand.

Eaton Corporation plc (ETN)

5 Year Price →

313.05
(+207.03%)



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Data [via](#) Yahoo! Finance

Opportunities in the power management space for data centers are growing for investors, including in ETFs like the First Trust NASDAQ Clean Edge Smart Grid Infrastructure Index Fund (GRID).

First Trust NASDAQ Clean Edge Smart Grid Infrastructure Index Fund (GRID)

5 Year Price →

123.07
(+120.11%)



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Data [via](#) Yahoo! Finance

Nuclear

Amazon, Microsoft, and Google are increasingly turning to nuclear energy to power their data centers.

And it's not just businesses - President Trump's treasury secretary announced in January that the department would support more nuclear reactors.

We sat down with Dr. Pavel Tsvetkov, a professor of nuclear engineering at Texas A&M, for an exclusive interview on the viability of nuclear energy for data centers.

Dr. Tsvetkov gave one simple reason as to why nuclear energy for data centers is growing:

“As we transition from fossil fuel, and we ultimately will transition, just simply because fossil fuel eventually will run out. Nuclear power has the ability to be sustainable.”

- Google signed a deal with Kairos Power in November 2024 to create a small modular reactor or SMR.

What's the big deal? SMR nuclear power can help data centers generate their own power, so they don't need to use power from the nation's grid.

Dr. Tsvetkov also believes that SMRs may be a game changer for powering data centers with nuclear energy in the future:

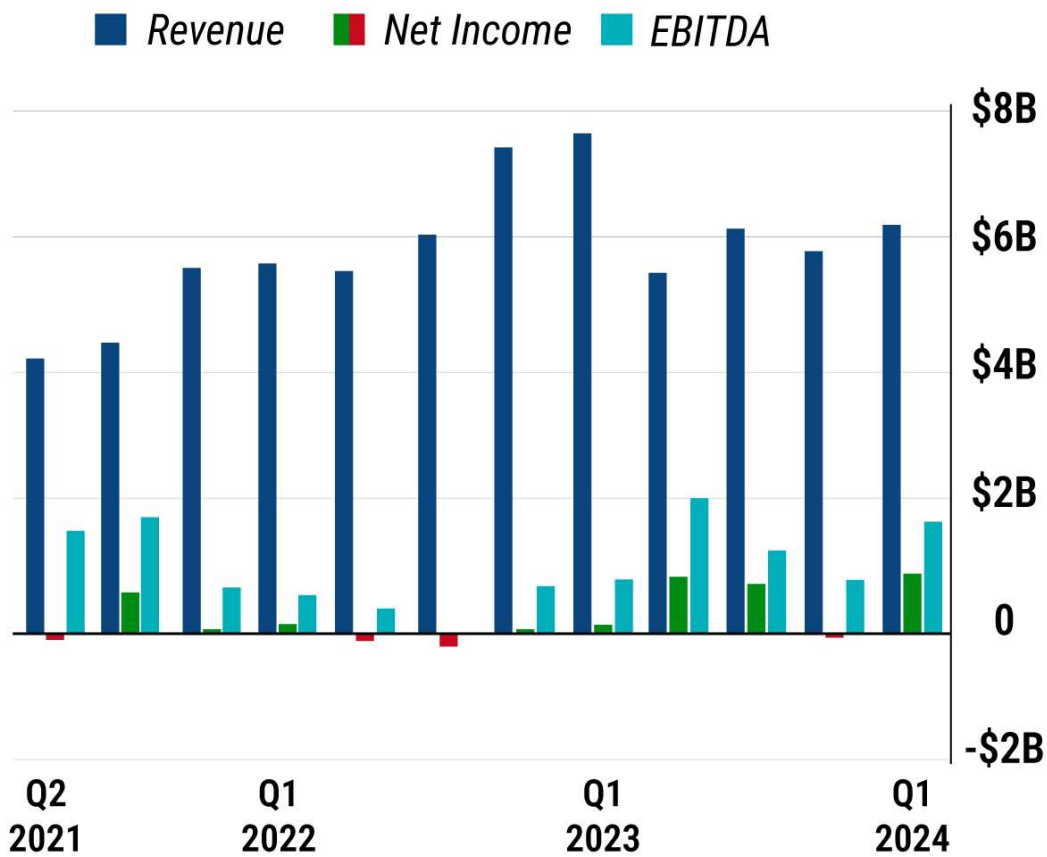
“The premise of small modular reactors is to take some of those challenges away by reducing the footprint of a nuclear power plant. We would have easier licensing requirements. We would have easier deployment requirements. It could be scalable.”

While Kairos Power is private, Constellation Energy (CEG) is one of the biggest public names in nuclear - and its also now powering data centers with SMRs.

Last September, Microsoft signed a 20-year deal with the Three Mile Island nuclear reactor to be restarted to power its various data centers.

Constellation Energy is working with the reactor to convert it to an SMR, which is located in Pennsylvania.

Constellation Energy Corp (CEG)



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Data via Yahoo! Finance

Note: Constellation Energy does not have any currently functioning SMRs, but it is the largest operator of nuclear power plants in the country.

- The company owns 21 reactors nationwide.

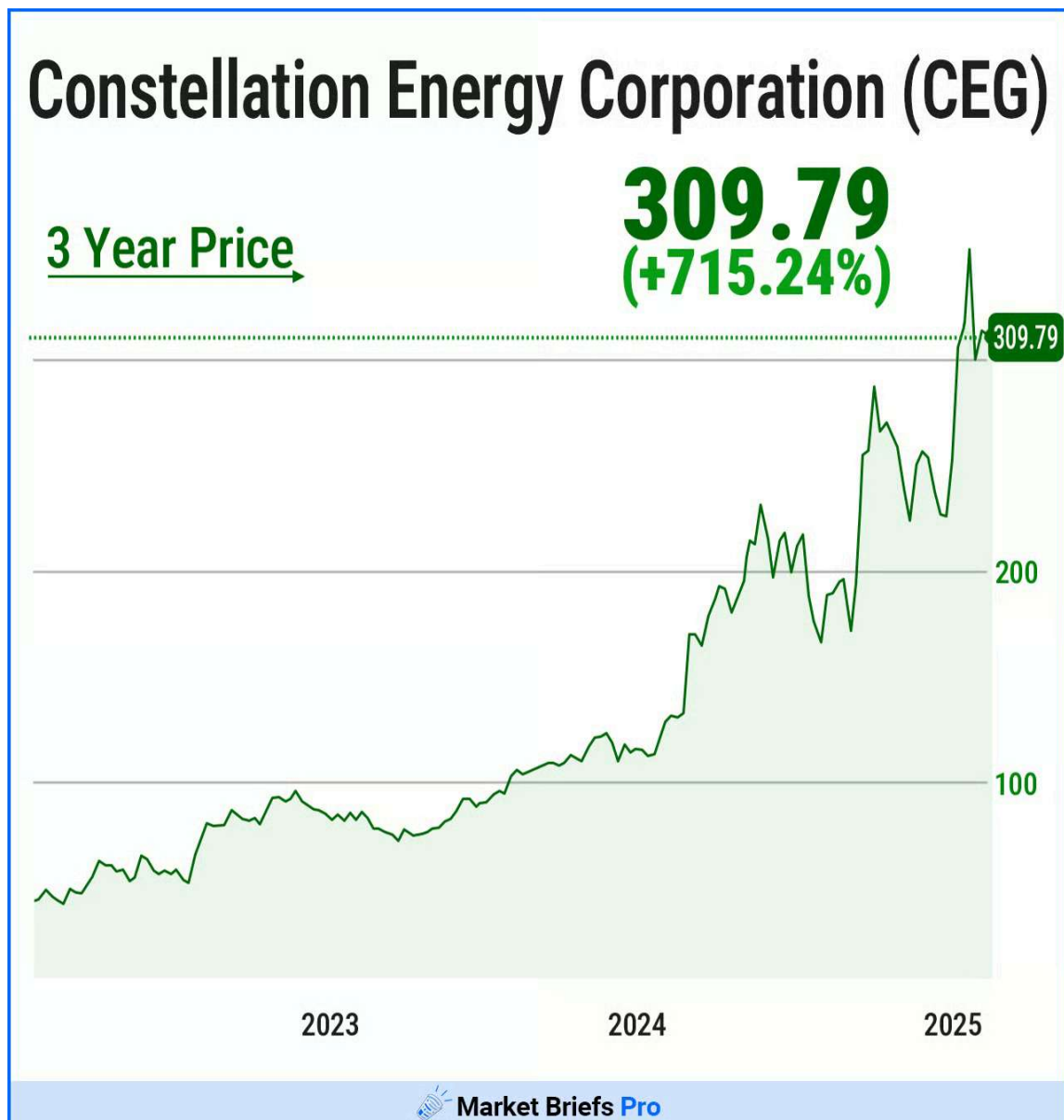
Outside of SMRs, the company is already powering data centers with its current reactors - like Microsoft's data center in Boydton, Virginia.

It signed a \$1 billion contract with the federal government in January, which will help it power over 1 million more homes with its nuclear energy by 2028.

Constellation has also been supplying five other data centers with over 2,500 megawatts of nuclear power since 2015 - making it the longest-running nuclear power provider in the data center industry.

As demand grows for data center power, so will its need for sustainable power, like from nuclear, creating more opportunities for Constellation

Energy.



Data [via](#) Yahoo! Finance

While Constellation Energy dominates the traditional reactor space, NuScale Corporation (SMR) has been prominent in the SMR space for years.

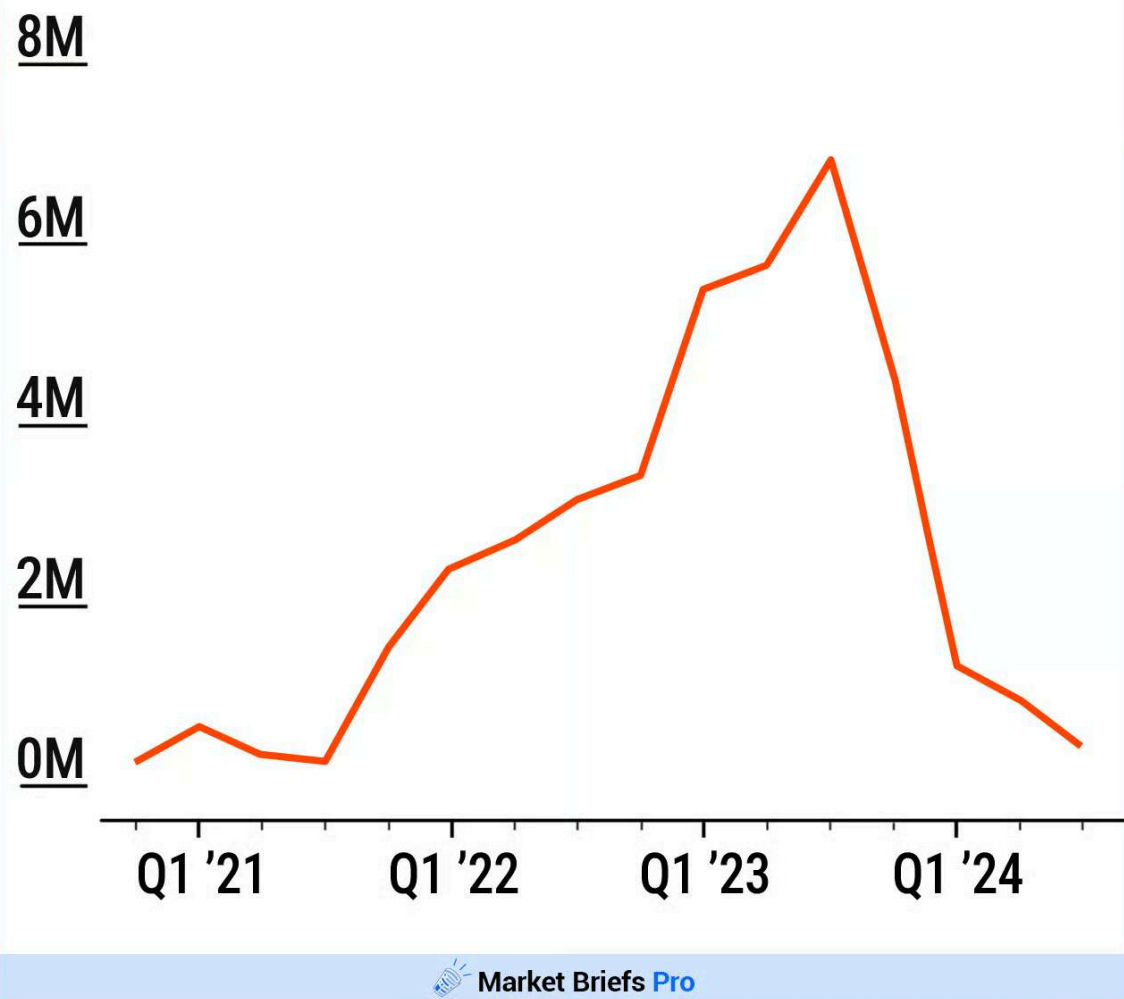
In 2023, NuScale signed a deal with Enbridge, to power its facilities with SMR tech.

Two power plants will primarily energize data centers in Ohio and are planned to be completed by 2029.

What else? SMRs from NuScale are already in use in Pennsylvania as well.

Those are the only locations NuScale has expanded into - but those contracts alone brought the company \$1.5 billion in revenue in 2024.

NuScale Revenue Growth And Decline Over Time



Data via Yahoo! Finance

It also holds the only patent to build a 50-megawatt SMR factory, which could power 50 data centers once built.

Currently, NuScale has taken on huge losses due to the cost of these reactors - it lost \$45 million in 2024 and \$58 million in 2023.

But its revenue could turn around once the SMR facilities are fully operational.

And as a major player in the SMR space for data centers, it's gaining market share without many other major competitors.

NuScale Power Corporation (SMR)

3 Year Price →

25.84
(+157.11%)



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Data [via](#) Yahoo! Finance

Constellation Energy and NuScale are two growing opportunities for investors in the nuclear energy data center space.

VanEck Uranium And Nuclear ETF (NLR)

5 Year Price →

87.73
(+78.20%)



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Data [via](#) Yahoo! Finance

General exposure to the nuclear energy space can also be found in the VanEck Uranium and Nuclear ETF (NLR).

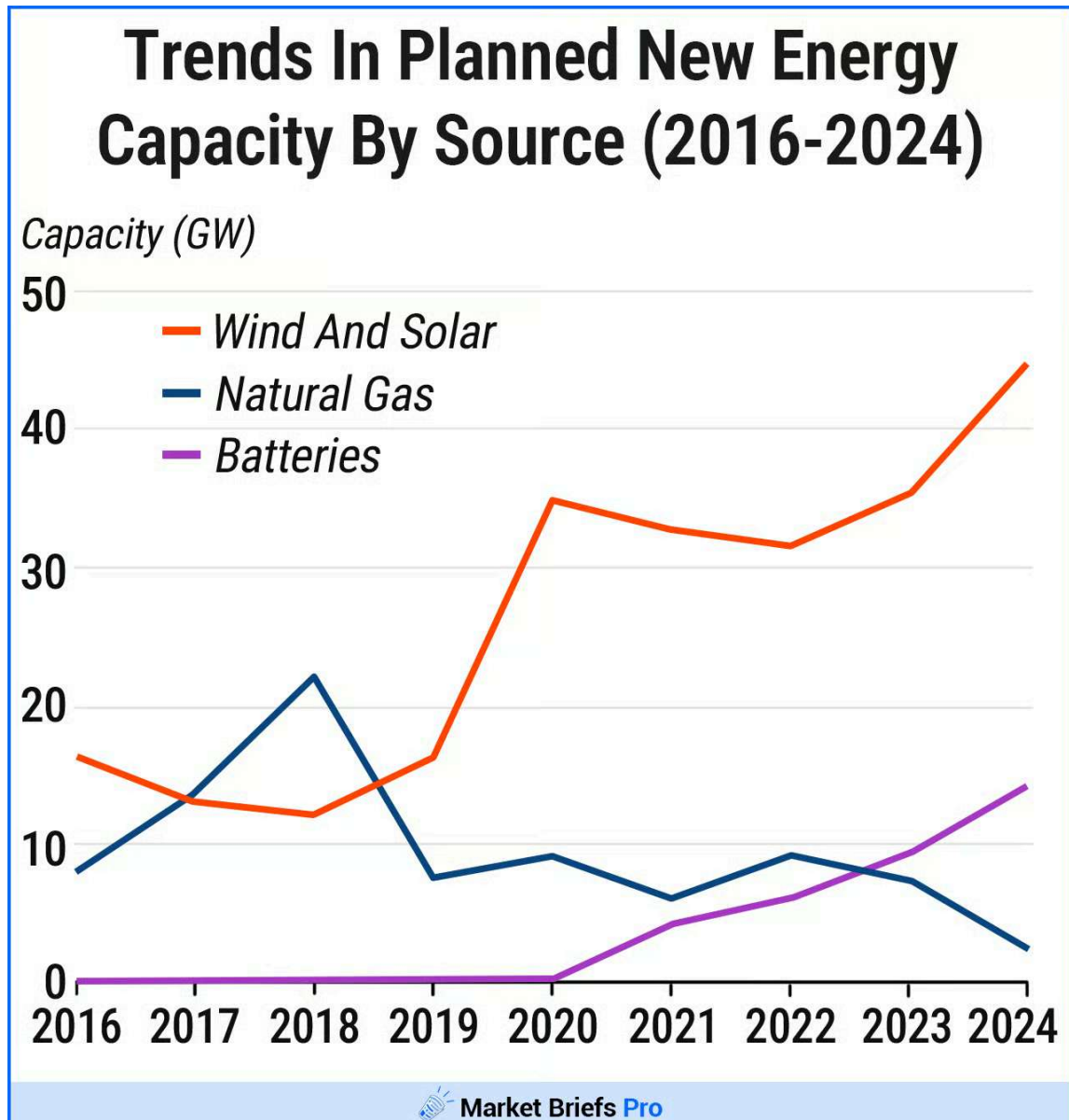
DEEP BRIEF

Wind

According to S&P Global, corporate spending on wind energy for data centers jumped by 50% year over year in 2023 - its largest annual increase ever.

Why? Research from Yahoo! found that wind energy could supply 92% of an entire data center's power by 2030 with just a few wind turbines.

Five tech companies - Meta, Apple, Microsoft, Google, and Amazon - plan on investing a combined \$100 billion into wind energy for their data centers by 2045.



Data [via](#) The White House

Some of that investment has already started.

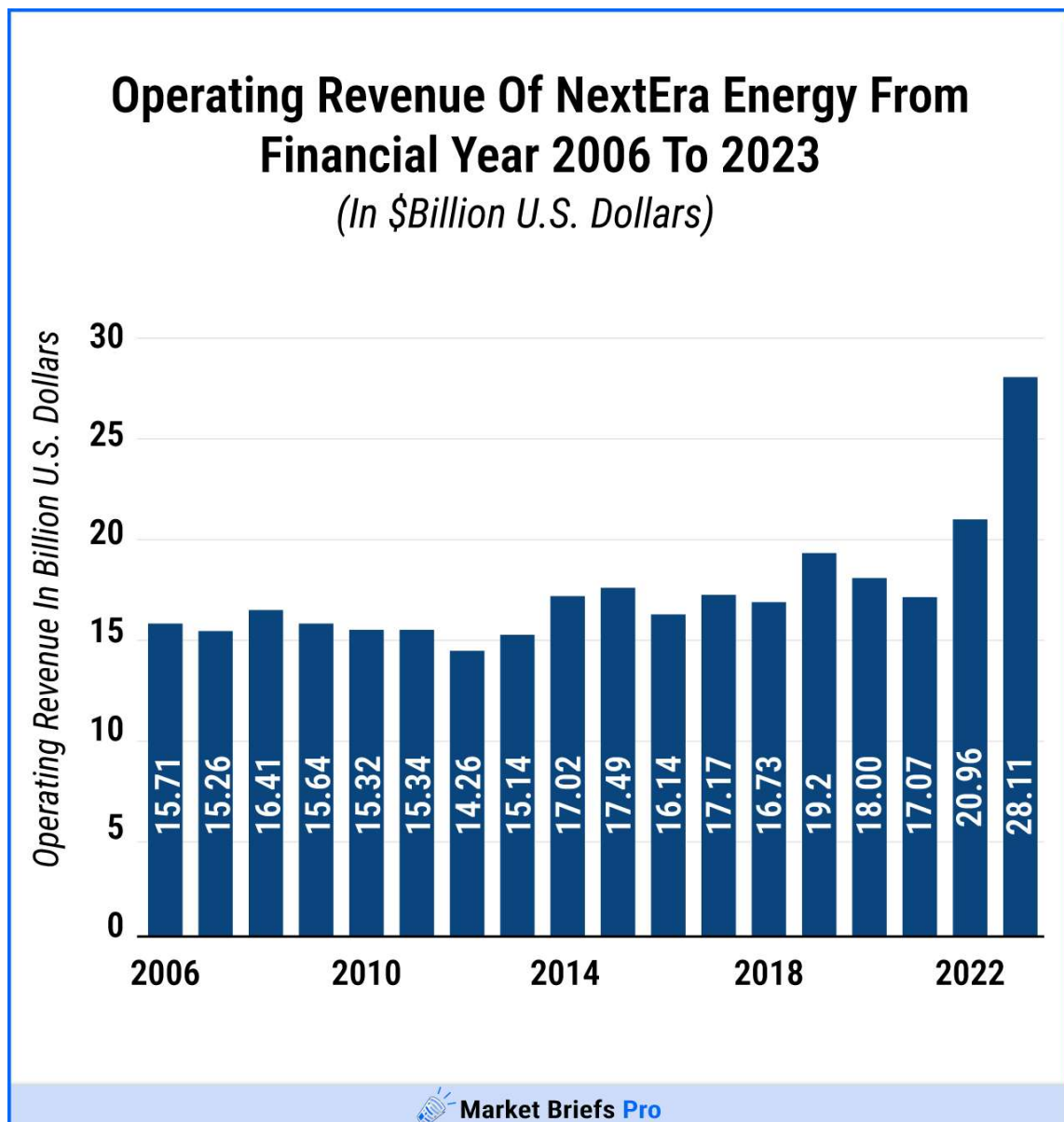
- In 2024, Google announced a \$1 billion deal in Scotland to power a data center using only wind energy.

All for wind? One wind turbine creates the same amount of energy as 50,000 solar panels.

But not so fast - President Trump signed an executive order on January 20th that ceases the use of all offshore wind projects for electricity.

- He's also ordered the Interior Department to audit the use of all current federal leases of wind power in the U.S.

Still, NextEra (NEE) Energy for example is currently on contract to supply power to data centers throughout the U.S.



Data via Statista

According to NextEra's latest earnings call, its data center powering contracts tripled in 2024, alongside AI and data center demand.

Most notably, the company inked a deal with Google to power one of its data centers by 2026.

Why NextEra? The company is already the largest producer of wind and solar power in the U.S.

More specifically - it holds the largest market share of wind power in North America, so most of its infrastructure is already in place and ready to be applied to data centers.

NextEra also brought in \$26 billion in annual revenue last year, an increase from just over \$25 billion in 2023.

NextEra Energy, Inc. (NEE)

68.27
(+1.31%)

5 Year Price →

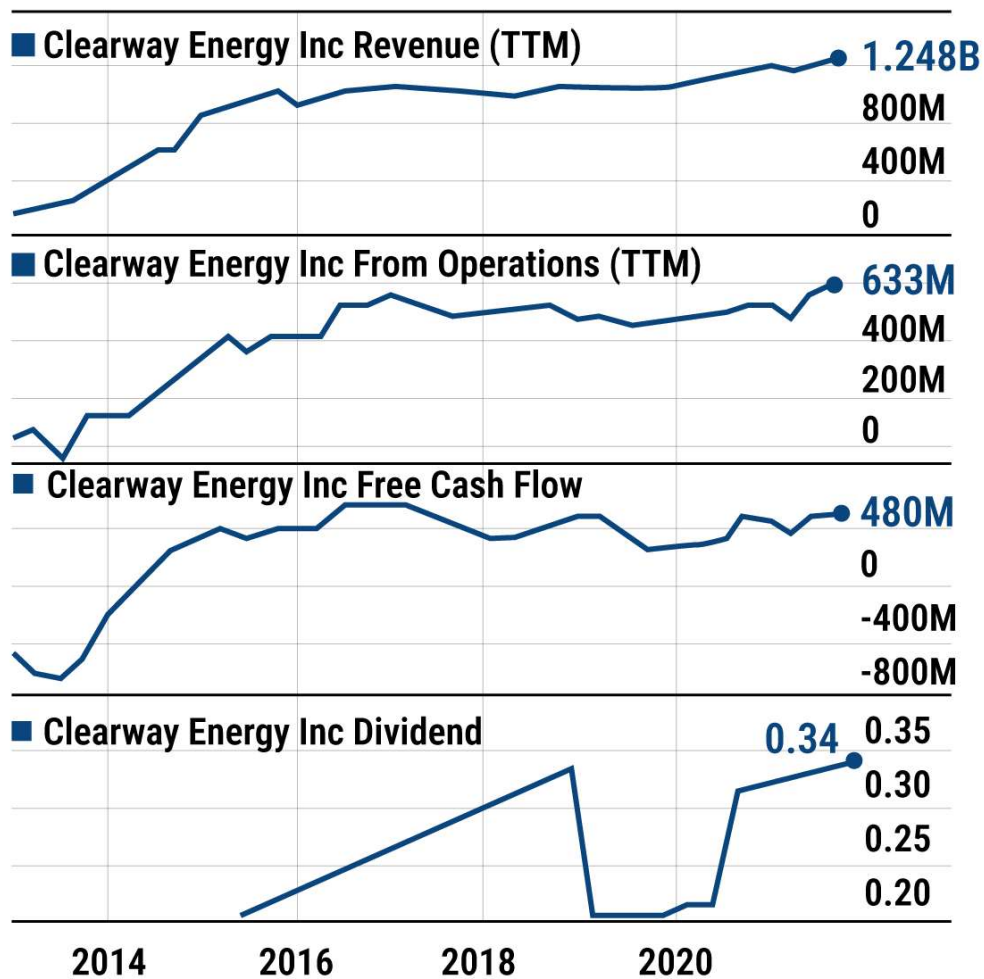


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Data [via](#) Yahoo! Finance

But NextEra isn't the only company capitalizing on wind energy for data centers - Clearway Energy (CWEN) specializes in connecting wind energy to data centers via the grid.

Clearway Energy Inc: Trends In Revenue, Cash Flow, And Dividends Over Time



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Data [via](#) The Motley Fool

What Clearway Energy specializes in though is storing energy and connecting data centers to the grid.

If a data center uses less energy than anticipated, it stores that extra energy for later - reducing costs and downtime for data centers.

In its latest earnings call, Clearway reported that it currently has five gigawatts of wind energy currently under contract for data centers throughout the U.S.

Clearway Energy, Inc. (CWEN)

5 Year Price →

26.43
(+22.33%)



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Data [via](#) Yahoo! Finance

- It added 3.6 gigawatts of wind power for data centers to its portfolio alone last year.

The company has also increased its revenue every year since 2022, with it reaching \$1.36 billion in 2024, a record high.

Data center demand + companies like NextEra and Clearway Energy = opportunities for investors to profit, along with funds like the SPDR S&P Kensho Clean Power ETF (CNRG).

SPDR Kensho Clean Power ETF (CNRG)

5 Year Price →

60.80
(+23.07%)



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Data [via](#) Yahoo! Finance

Risks

While energy for data centers is increasing, investors should be aware of a few things:

First, The U.S. has only opened on average five nuclear power plants a year since 2010 - down from the near 15 a year in 1975.

Why? Congress is largely split on its regulation, and nuclear plants can be expensive compared to traditional coal or natural gas resources for power.

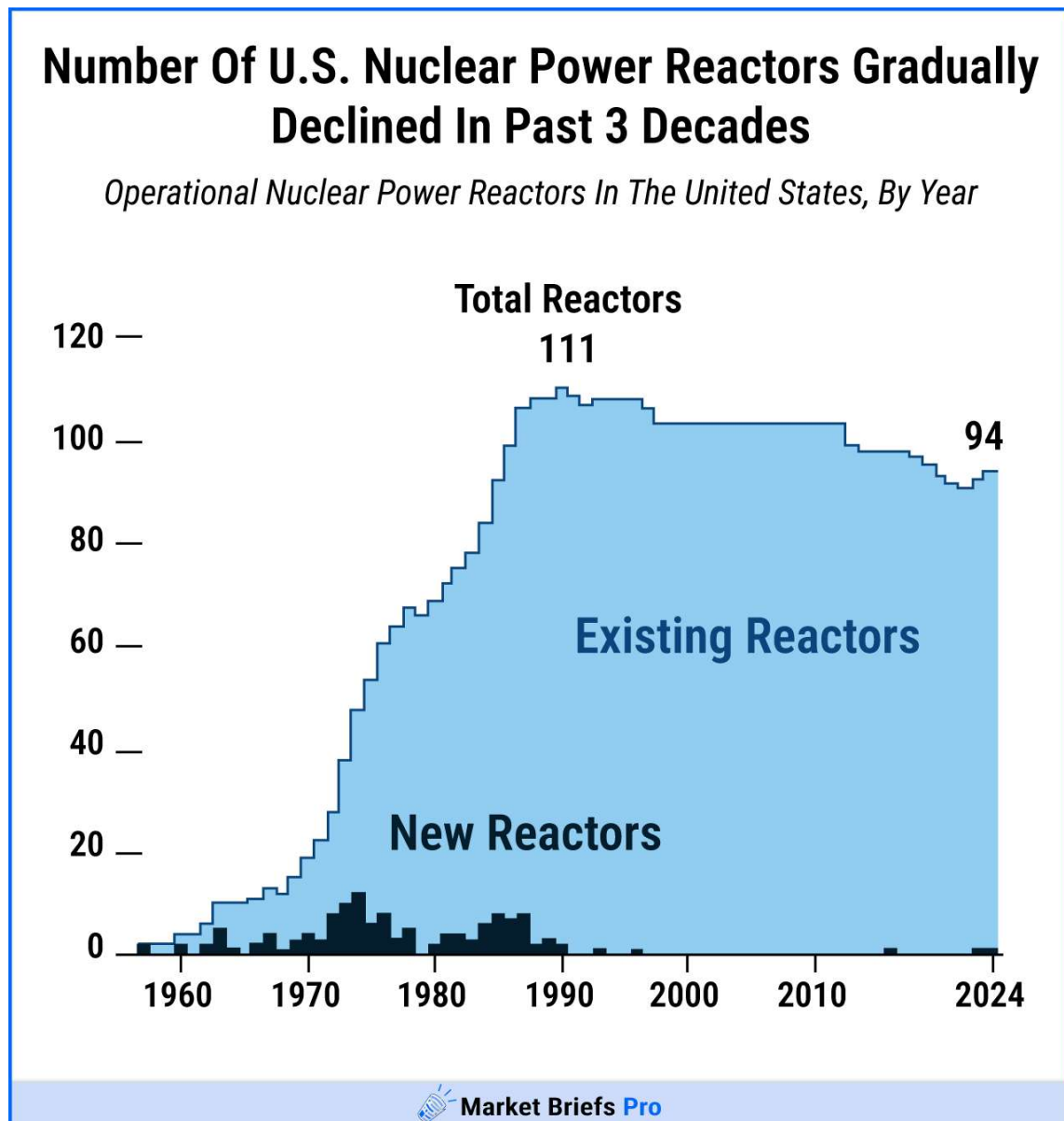
- Other renewable energy sources like wind and solar also typically see less funding and federal legislative support under a Republican president.

Dr. Tsvetkov attributes this to another reason: Economics.

"It doesn't really matter for a company whether it's a nuclear power plant or gas plant. If nuclear is commercially viable and the economics works then they will operate it."

Second, data center demand is high today, but there is no guarantee it will remain this high forever.

It could be decades before the contracts for these power deals are completed and demand could look very different in the future.



Data via Pew Research

Third, not only do these power solutions take time to build, they also impact a business's revenue.

Reminder: NuScale Energy saw its revenue drop by 93% in 2024, its second straight year of revenue decline.

Lastly, the release of DeepSeek, (an AI similar to ChatGPT), from China caused U.S. power and data center stocks to fall by more than 20% in late

January.

Why? DeepSeek data is routed through China - meaning it could take market share from ChatGPT, resulting in less energy usage by an American company.

- In January 2025, downloads of DeepSeek on the Google Play and Apple App store exceeded ChatGPT.

REVIEW

Energizing Data Centers

Data centers will need to generate more power to meet demand.

That's because without data, businesses won't have access to the power they need to create new innovations in AI, quantum computing, and more.

For nuclear energy and data centers, we are seeing growth, according to Dr. Tsvetkov:

"We have probably more than 50 different companies pursuing different forms of nuclear power plants...so it's quite dynamic."

Power solution companies are working on meeting that demand, with many making a large impact on the space already, with more to come.



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